R13

Code No: 117AN

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year I Semester Examinations, January/February - 2023
ARTIFICIAL INTELLIGENCE
(Computer Science and Engineering)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

PART – A

(25 Marks)

1.a)	List out the application of AI.	[2]
b)	Explain characteristics of a problem.	[3]
c)	What are the components of Propositional Logic?	[2]
d)	Define a frame with example.	[3]
e)	Define Hybrid expert system.	[2]
f)	Explain explicit representation.	[3]
g)	Distinguish between knowledge and learning.	[2]
h)	Distinguish supervised and unsupervised learning.	[3]
i)	Define Transition Network Parser.	[2]
j)	What are Augmented Transition Network Parsers?	[3]

PART – B

(50 Marks)

2. Represent 8 puzzle problem in state space. And apply best first search to solve. [10]

OR

3. Explain the challenges and applications of artificial intelligence. [10]

4. Express the following sentences as semantics Net structure and conceptual dependencies.
a) All those who code in C language are a programmer
b) Sam gave Mary a box of Candy. [5+5]

OR

5. Explain predicate calculus. Discuss inference rules to produce calculus expression. [10]

6. What is unification? Explain Logic based abductive Inference. [10]

OR

7. Explain the process of knowledge acquisition and validation for expert systems. [10]

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8. Explain ID3 algorithm and its role in building a decision tree. Give example. [10]

OR

9. Define Unsupervised Learning and discuss in detail about Conceptual Clustering. [10]

10.a) Explain the role of knowledge in language understanding.
b) Explain various applications of Natural Language processing. [5+5]

OR

11. Give examples to explain the concept of sentence analysis phase of natural language processing. [10]

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